Cover Letter to the Oregon State University HRPP / IRB

To: Oregon State University Human Research Protection Program (HRPP) / Institutional Review Board
From: Dr. YanMing Di, Professor, Department of Statistics, Oregon State University
Re: New IRB application — *Passive Forearm sEMG/IMU Monitoring During Normal Tennis Practice: A Minimal-Risk Two-Participant Pilot Case Series in Minor Student-Athletes*
Date: [submission date TBD]

Dear OSU HRPP / IRB,

I am submitting a new application for a minimal-risk pilot case-series study using passive wearable surface electromyography (sEMG) and inertial measurement unit (IMU) sensors during ordinary tennis practice. The two participants are minor student-athletes who are also student co-authors on the engineering manuscript that this work supports. The study collects sEMG, IMU, and simple subjective fatigue/exertion/pain ratings; it collects no video, no audio, no heart-rate, no SpO_2, and no other clinical measures, and it does not direct the participants to exceed their normal tennis activity.

The study is designed as **non-exempt minimal-risk research** under 45 CFR 46.404 and is anticipated eligible for **expedited review** at 45 CFR 46.110 Category 7 (research on individual or group behavior, characteristics, or non-invasive procedures). It is conducted in conformance with the U.S. Common Rule, 45 CFR 46 Subpart D, and the Declaration of Helsinki.

The application explicitly addresses an unusual study-team structure that the IRB should be aware of:

1. One minor participant (Theodore Di) is the son of the principal investigator.
2. The other minor participant (Ryan Wang) is the son of the technical assistant (Xiaodong Wang), who provides on-court technical support and equipment assistance.
3. Both minors are also intended student co-authors on the resulting engineering manuscript.

This creates symmetric undue-influence and conflict-of-interest concerns. The protocol mitigates these through:

- **An Independent Assent Administrator** (Doc 8) — an OSU faculty/staff designee who is not a parent, not a co-author, and has no publication interest in the study. The IAA conducts or witnesses each minor's assent discussion privately and confirms voluntariness.
- **A written Conflict of Interest Management Plan** (Doc 6) accepted before any data collection.

- **Participation–authorship separation**: a minor's withdrawal from data collection does not affect family relationships, tennis opportunities, school standing, or appropriate scholarly credit for already-completed work.
- **Two-parent permission when feasible** (although §46.408(b) allows the IRB to find one parent sufficient for §46.404 work, this protocol elects to seek both parents' written permission for both minors as additional protection).
- **CITI Group 1 or Group 2 — Human Subjects Protection** completion by all study team members before submission, documented in Appendix D.

The study uses **personally-owned hardware** and does not draw on OSU laboratory facilities, equipment, or budget. OSU's role is institutional review and PI oversight. Data collection occurs at two private residential tennis courts: one in [city], California (Ryan), and one in Corvallis, Oregon (Theodore). Neither high school is involved as a research site.

The work is intended for submission to an IEEE biomedical venue (e.g., IEEE EMBC, IEEE JBHI, IEEE JTEHM, IEEE TBME, or IEEE Sensors). The manuscript will include the IEEE-required IRB-approval reference, parental-permission and minor-assent statements, Helsinki conformance, conflict-of-interest declaration, and AI-disclosure paragraph (the latter only if AI-assisted drafting is actually used).

I and the study team are available for any clarification or modification the IRB requests. Please direct correspondence to me at the email above.

Respectfully,

Dr. YanMing Di
Professor, Department of Statistics
Oregon State University
Corvallis, OR 97331
Yanming.Di@oregonstate.edu

Document 1 — IRB Protocol

1. Study Title

Passive Forearm sEMG/IMU Monitoring During Normal Tennis Practice: A Minimal-Risk Two-Participant Pilot Case Series in Minor Student-Athletes.

2. Study Team and Roles

Name	Role	Affiliation	Notes
Dr. YanMing Di	**Principal Investigator** (faculty PI)	Department of Statistics, Oregon State University	Independent oversight, IRB submission, statistical and scientific supervision. Guided sensor hardware/software design, contributed to algorithm discussions, and applied mathematical modeling to EMG/IMU signal analysis. Also the parent of minor participant Theodore Di — disclosed COI; mitigated per Doc 6.
Ryan Wang	Student researcher, lead student co-author, sensor participant (minor)	Westview High School	Designed the sensor hardware/software; primary data collection at the California site; analysis pipeline; manuscript drafting.
Theodore (Teddy) Di	Student researcher, student co-author, sensor participant (minor)	Crescent Valley High School	Designed the sensor hardware/software; second-site data collection (Corvallis); cross-site reproducibility analysis; manuscript review.
Xiaodong Wang	**Technical assistant** (not an author)	Independent	Provides on-court technical support, sensor setup assistance, and equipment custody/maintenance. Will appear in manuscript Acknowledgments. Also the parent of minor participant Ryan Wang — disclosed COI; mitigated per Doc 6.
[TBD]	**Independent Assent Administrator** (IAA)	OSU faculty/staff designee	Conducts/witnesses each minor's assent discussion privately. Must not be a parent, co-author, technical contributor, supervisor, or anyone with publication interest in the study. See Doc 8.

Role clarification. The faculty PI provides independent oversight and is responsible for the IRB submission and study conduct. The technical assistant provides hardware/software and on-court

support, including safety supervision at the California site. Each minor's assent discussion is conducted or supervised by the IAA, who is structurally outside the family/coauthor relationships of the study team.

3. Study Purpose

The purpose of this study is to evaluate the feasibility and field validity of passive wearable forearm sEMG and IMU monitoring during normal tennis practice in two minor student-athletes who are also student researcher/co-authors on the project. The study examines whether forearm sEMG features and motion context can characterize fatigue-related trends during ordinary tennis activity, and documents field-deployment factors such as electrode settling, contact quality, motion artifacts, and session continuity across two distinct residential courts.

The study is **not** intended to diagnose, treat, prevent, or predict any medical condition or injury, and the system is **not** a medical device.

4. Study Design

This is a prospective, minimal-risk, two-participant pilot **case series** under 45 CFR 46.404. Only data collected after IRB approval will be used for research analysis, conference submission, or publication.

Each participant will complete approximately **5 to 10 sessions**, each up to approximately **90 minutes** (including setup, warmup, normal tennis practice or practice-match activity, brief subjective ratings, and sensor removal). The study does not require exercise beyond the participant's usual tennis activity. Participants are not instructed to play to exhaustion or to perform maximal-effort fatigue protocols. Rest breaks may be taken at any time.

The two-site, two-participant design is motivated scientifically by the need for **cross-site reproducibility** of the field-validity claims developed in the technical manuscript: a single residential court could leave open whether observed sEMG settling/fatigue trends are court- or athlete-specific. A second site with a second player provides a defensible second observation.

5. Study Locations

Data collection occurs at two private residential tennis courts. Neither participant's high school (Westview HS or Crescent Valley HS) is a research site; the high schools are listed for affiliation only. There is no school-based recruitment.

Site A — California residential court (Ryan Wang):

- Location: residential tennis court in [city], California [exact address withheld from this packet but provided to the IRB on request].
- Adult presence: Xiaodong Wang (technical assistant and parent) is on-court or courtside throughout each session.
- No public spectators are invited by the research team. No school or team involvement.

Site B — Corvallis, Oregon residential court (Theodore Di):

- Location: residential tennis court in Corvallis, Oregon [exact address withheld from this packet but provided to the IRB on request].
- Adult presence: Dr. YanMing Di (PI and parent) or another approved adult is on-court or courtside throughout each session.
- No public spectators are invited by the research team. No school or team involvement.

Sessions at either site are not conducted during unsafe weather, poor lighting, wet court conditions, or other unsafe conditions.

6. Participant Population

Number of participants: **two**.

Participant	Age at start	Status
Ryan Wang	17	Turns 18 in September 2026. If data collection continues after his 18th birthday, he will be asked to provide written adult informed consent before any post-18 collection.
Theodore (Teddy) Di	15	Turns 16 in September 2026; turns 18 in September 2028. Adult re-consent provision applies if data collection continues past his 18th birthday.

Both participants are minor tennis players in regular tennis practice. Both are also student researcher/co-authors on the engineering manuscript supported by this study.

7. Inclusion Criteria

A participant must:

1. Be one of the two named minor tennis players described above.
2. Participate in regular tennis practice at the named residential court.

3. Be able to understand the study procedures and provide written assent.
4. Have written parental permission (preferably from both parents — see §10).
5. Be willing to wear passive forearm sEMG/IMU sensors during normal practice sessions.

8. Exclusion / Pause Criteria

A session will not be conducted, or will be paused, if any of the following apply on a given day:

1. Current wrist, forearm, elbow, shoulder, or hand injury or pain that affects tennis participation.
2. Known allergy or sensitivity to adhesive electrodes, tape, or sensor straps.
3. Open wound, rash, irritation, or skin condition at the planned sensor placement location.
4. The participant does not wish to participate or requests to stop.
5. The on-court adult or the IAA determines participation is not appropriate on a given day.
6. Unsafe weather, wet court, poor lighting, or other unsafe court conditions.

9. Recruitment and Enrollment

There is **no public recruitment** and **no recruitment of additional minors**. The study is a closed cohort limited to the two named participants who are also student researcher/co-authors on the project. Enrollment occurs only after IRB approval, written parental permission, and written minor assent are obtained.

10. Consent and Assent Procedures

Because both participants are minors, the study obtains **written parental permission** and **written minor assent** before data collection.

Permission (parental):

- Per §46.408(b), the IRB may find that permission of one parent is sufficient for research approvable under §46.404. The protocol nevertheless **elects to seek written permission from both parents of each minor when feasible** as additional protection against perceived undue influence given the close family relationships within the study team. If only one parent's permission is obtained, the application will document the regulatory basis and the IRB's accepted justification.
- Permission forms are adapted from OSU's current "Consent Form template for non-exempt studies" (see Doc 2A and 2B).

Assent (minor):

- Required from each minor under §46.408(a). Both Ryan (17) and Theodore (15) are capable of providing meaningful written assent.
- The assent discussion is **conducted or witnessed privately by the Independent Assent Administrator (IAA)** — not by the parent, the PI, the technical assistant, or any co-author.
- The IAA confirms with each minor that participation is voluntary, may be declined or withdrawn at any time without penalty, and is not required to maintain family relationships, tennis opportunities, school standing, or scholarly credit/authorship for work already completed.
- The assent forms (Doc 3A and 3B) use OSU's current "Assent Form" template, written in plain language appropriate for a 16- or 17-year-old mature minor.

Participation–authorship separation: authorship and acknowledgment decisions for any future manuscript are not contingent on a minor's continued data-collection participation. A participant may withdraw from data collection while still receiving appropriate scholarly credit for work already completed, where applicable under publication standards.

11. Study Procedures

11.1 Pre-session safety check

At the start of each session, the on-court adult and (where applicable) the IAA confirm that:

- The participant still agrees to participate.
- The participant has no pain, injury, skin irritation, or feeling that would make tennis participation inadvisable.
- Court and weather conditions are safe.
- The participant is reminded that he may rest or stop at any time.

11.2 Sensor placement

A passive surface EMG sensor (MyoWare 2.0) is placed on the skin over the forearm muscle region involved in tennis strokes — typically the wrist flexor and/or wrist extensor region. A reference/ground electrode is placed at an appropriate bony landmark (e.g., near the lateral epicondyle), consistent with vendor guidance and PI review. A passive IMU motion sensor (ICM-20948) is secured on the forearm using a strap or other non-invasive support.

The sensor kit is **passive only**. It records naturally occurring muscle electrical activity and motion. It does **not** stimulate the muscle, deliver medication, deliver heat, or provide treatment. See §32 (Device Safety Statement) and Doc 5 for the full device description.

11.3 Baseline / warmup recording

A short baseline recording may be collected at rest and during light movements (e.g., gentle wrist motion, racket holding, warmup strokes). This baseline supports signal-quality assessment and later analysis only; it is not a medical evaluation.

11.4 Tennis activity recording

The participant performs normal tennis practice activities or practice matches. Practice may include rallying, serves, forehands, backhands, volleys, footwork, and ordinary match play. The study does not prescribe additional training load beyond regular practice.

11.5 Subjective ratings

At the end of each session — or at selected time points if approved by the IRB — the participant may be asked to provide simple subjective ratings on 0–10 scales:

- **Forearm fatigue:** 0 = none, 10 = maximum.
- **Overall exertion:** 0 = none, 10 = maximum.
- **Pain/discomfort:** 0 = none, 10 = maximum.

If the participant reports pain or unusual discomfort, the session is stopped and the parent is notified if not already present.

11.6 Sensor removal and skin check

After activity, sensors are removed. The participant is asked whether there is skin irritation, discomfort, or other concern. If irritation occurs, the site is monitored and the parent is informed.

12. Data Collected

- Forearm sEMG signals from passive surface electrodes (MyoWare 2.0 ENV + RAW outputs).
- Forearm motion data from a passive IMU (accelerometer/gyroscope; ICM-20948).
- Session timing and activity labels (e.g., practice, rally, serve, practice match), if recorded.
- Subjective fatigue, exertion, and pain/discomfort ratings on 0–10 scales.
- Minimal contextual information needed for analysis: session date, approximate duration, dominant arm, general tennis context.

The study **does not** collect heart rate, SpO_2, blood pressure, ECG, or any other vital-sign data; **does not** record video or audio; and **does not** collect biospecimens.

13. Data Analysis Plan

The research team analyzes sEMG features such as RMS amplitude, median frequency (MDF), mean frequency (MNF), and signal quality / contact-related metrics. IMU data is used to identify swings, estimate movement intensity, and provide motion context for interpreting EMG changes. Subjective ratings may be compared with sensor-derived features. Analyses are descriptive and exploratory because the study involves only two participants.

The study **avoids clinical claims**. Results are described as feasibility, field-validity, signal-quality, or fatigue-consistent observations — not diagnosis, treatment, or injury prediction.

14. Risks and Mitigation

The expected overall risk level is **minimal**. The main risks are mild skin irritation, ordinary tennis fatigue/soreness, privacy or identifiability risk, and potential undue influence due to the family/co-authorship structure of the study team. A detailed risk assessment is provided in **Doc 4 — Risk Assessment**; mitigations are summarized there and in §10, §16, §22, and §23.

14.1 Ordinary tennis risk vs. research-added risk

The IRB should distinguish ordinary tennis risk (which exists with or without this study) from research-added risk (which exists because of this study).

- **Ordinary tennis risk** includes muscle soreness, fatigue, falls, and minor injury inherent to tennis itself. The research does **not** require participants to exceed their normal practice intensity.
- **Research-added risk** is limited to: possible mild skin irritation from adhesive electrodes/tape/straps; minor inconvenience from sensor wear; and confidentiality/identifiability risk because this is a small named cohort.

15. Benefits

There is **no guaranteed direct benefit** to either participant. Possible indirect benefits include educational experience in wearable sensing, signal processing, and responsible human-subjects research. The study may contribute to improved understanding of field-deployable sports sensing methods.

16. Privacy, Confidentiality, and Identifiability

Sensor files are labeled using a participant code rather than the participant's name. The key linking each participant's identity to the code is stored separately from the sensor data. Data is stored on a

password-protected, encrypted local computer or OSU-licensed storage controlled by the research team. Access is limited to the named study team.

Because the study is a two-participant case series and the participants are also intended co-authors, **complete anonymity cannot be guaranteed**. Publications will avoid unnecessary personal information and will not include identifiable images, video, or audio. This limitation is disclosed in the parental permission and minor assent forms.

Off-site privacy controls:

- No public spectators are invited by the research team.
- No video or audio is recorded.
- No school or team involvement; collection occurs at private residential courts only.
- A supervising adult is present for safety; collection is postponed for unsafe weather or court conditions.

17. Data Storage, Retention, and Sharing

- **Storage:** raw and analyzed data stored on encrypted local drives or OSU-licensed storage; identity-key file held separately on encrypted local storage; routine encrypted backup; patched OS; firewall; access limited to the named study team.
- **Retention:** coded research data retained for **[N] years** after publication or study closure (e.g., 5 years; final value to be set by the PI before submission). The identity-key file may be destroyed earlier when no longer needed. De-identified aggregate summary metrics may be retained longer if the IRB approves.
- **Sharing:** raw data are **not** publicly shared under this protocol. De-identified summary metrics may be shared with an IEEE journal or conference reviewer or made available to the corresponding author on reasonable request, consistent with the publication's data-availability requirements.
- **Future use:** data are used only for the approved study and the related publications/presentations described in this IRB application. Materially different future use will require an IRB amendment.

A full data security plan is provided in **Doc 7 — Data Security Plan**.

18. Withdrawal

Each participant may stop any data collection session at any time, may decline a session, and may request that data from an incomplete or uncomfortable session not be used. Once de-identified results have been analyzed, submitted, accepted, or published, complete withdrawal of already-incorporated data may not be possible. This limitation is explained during parental permission and minor assent.

Withdrawal from data collection does **not** affect family relationships, tennis opportunities, school standing, or appropriate scholarly credit for work already completed.

19. Stopping Rules

Data collection for a session or for the study as a whole is stopped if any of the following occur:

- The participant requests to stop.
- Pain, unusual discomfort, dizziness, distress, or concerning fatigue occurs.
- Skin irritation or adhesive reaction occurs.
- Unsafe weather, wet court, poor lighting, or unsafe court conditions occur.
- The on-court adult, the PI, or the IAA determines that continuing is not appropriate.

20. Emergency / Safety Plan

If injury, pain, dizziness, or distress occurs, the session is stopped immediately. The other parent is notified if not already present. The study team follows site safety procedures and seeks medical assistance if needed. Researchers do not provide medical diagnosis or treatment. The technical assistant carries a charged phone for emergency contact during California-site sessions; the PI does the same at the Oregon site.

21. Existing Data

Only data collected after IRB approval will be used for research analysis, conference submission, or publication. Any prior engineering test data — collected before IRB approval as part of system development — will not be used as human-subjects research data unless the IRB explicitly allows it through a written determination.

22. Conflict of Interest

The study team contains two structurally significant family relationships:

- The PI (Dr. YanMing Di) is the parent of minor participant Theodore Di.
- The technical assistant (Xiaodong Wang) is the parent of minor participant Ryan Wang.

Both minors are also intended student co-authors on the resulting engineering manuscript. This creates symmetric undue-influence risk and potential publication-interest bias.

A full **Conflict of Interest Management Plan** is provided in **Doc 6**. Key elements:

- Independent Assent Administrator conducts/witnesses each minor's assent privately (Doc 8).

- Participation–authorship separation: withdrawal does not erase scholarly credit for already-completed work.
- COI disclosure to OSU through the Beaver Research Hub before submission.
- COI declaration to be included in the eventual IEEE manuscript.

23. Independent Assent Administrator (IAA)

To address undue-influence risk in a closed-family study team, the protocol designates an **Independent Assent Administrator** — an OSU faculty/staff member who is not a parent, not a co-author, not the technical assistant, not a supervisor, and has no publication interest in the study.

The IAA:

- Conducts or witnesses the assent discussion with each minor privately (without the parent or PI present).
- Confirms that each minor understands participation is voluntary, may be declined or withdrawn at any time without penalty, and is not required to maintain family, athletic, authorship, or academic relationships.
- Signs the Person-Obtaining-Assent line on each minor assent form (Doc 3A and 3B).
- May halt a session under §19 if the minor appears pressured.

The IAA designation is documented in **Doc 8 — Independent Assent Administrator Designation**.

24. CITI Training

All study team members complete CITI **Group 1 or Group 2 — Human Subjects Protection** before submission, renewed every three years. Responsible Conduct of Research training is **not** a substitute for the Human Subjects Protection course at OSU.

The CITI Training Tracker is provided in **Appendix D**.

25. 18th-Birthday Re-Consent

If data collection continues for a participant past his 18th birthday, the protocol pauses new collection for that participant until written **adult informed consent** is obtained. Pre-18 data remain usable under the original parental permission and minor assent unless the now-adult participant withdraws as allowed under §18. If the now-adult participant declines re-consent, no further data collection occurs for him.

The PI submits an IRB amendment, if required, before the 18th-birthday transition for each participant.

26. Continuing Review and Amendments

Continuing review and amendments are managed through OSU Cayuse HE according to the approval letter timeline. Material changes to procedures, study team, sites, or risk profile are submitted as amendments before implementation.

27. Adverse Event and Deviation Reporting

Reportable events for this study include, but are not limited to:

- Skin redness lasting beyond a session.
- Rash or blistering from electrodes, tape, or straps.
- Pain or unusual discomfort from device placement.
- Privacy breach, lost device, or lost data.
- Distress or perceived pressure to participate.
- Any tennis injury during a research-recorded session — even if likely ordinary sport-related — is logged and evaluated.

Adverse events and protocol deviations are reported through OSU Cayuse HE per HRPP timelines. Full procedure is described in **Appendix C — Adverse Event and Continuing Review Procedure**.

28. Compliance Statements

This study is conducted in accordance with:

- the U.S. Common Rule, 45 CFR 46 (revised 2018);
- 45 CFR 46 Subpart D — Additional Protections for Children;
- the Declaration of Helsinki (current revision, World Medical Association); and
- the Oregon State University HRPP Policy and Procedure Manual.

29. Publication Plan

The study results may be submitted to an **IEEE biomedical venue** — for example, IEEE EMBC, IEEE Journal of Biomedical and Health Informatics (JBHI), IEEE Journal of Translational Engineering in Health and Medicine (JTEHM), IEEE Transactions on Biomedical Engineering (TBME), or IEEE Sensors. The manuscript will include:

- the IEEE-required statement that the research was performed under the oversight of the Oregon State University HRPP / IRB, including the protocol/approval number;
- the IEEE-required statement that written parental permission and written minor assent were obtained;

- a Declaration of Helsinki conformance statement;
- a conflict-of-interest declaration disclosing the parent–child relationships within the study team and the IAA mitigation;
- an author-contributions section meeting IEEE's authorship criteria; and
- an AI-disclosure paragraph in the Acknowledgments **only if** AI-assisted drafting is actually used in the manuscript.

The study is presented as a two-participant pilot case series and does **not** make clinical, diagnostic, or injury-prediction claims.

30. Compensation and Cost

There is **no compensation** for participation. Participants incur **no cost**. Participation does not affect school standing, tennis opportunities, family relationships, authorship expectations, or any other benefits.

31. Future Use of Data

Data collected under this protocol are used only for the approved study and the related publications/presentations described in this IRB application. Materially different future use — for example, broader biomechanics studies, machine-learning model training datasets distributed externally, or use beyond the IEEE manuscript and any directly derived presentations — will require an IRB amendment.

32. Device Safety Statement

The study system records signals only. Specifically:

- The MyoWare 2.0 surface EMG sensor records naturally occurring muscle electrical activity from the skin surface. It does **not** deliver electrical stimulation, medication, heat, or any therapeutic intervention.
- The ICM-20948 IMU records motion only. It does not stimulate or interact with the body in any way.
- The system is battery-powered. There is no mains-electrical connection during data collection.
- The system is used only for non-clinical research signal recording and engineering analysis. It is **not** used to diagnose, treat, prevent, or monitor a medical condition. **No clinical decisions** are made from the data.
- The participant may remove the sensors at any time. Sessions are stopped immediately for discomfort, skin irritation, dizziness, pain, or participant request.

A full sensor and device description is provided in **Doc 5 — Sensor and Device Description**.

Document 2 — Parental Permission Forms

OSU's current "Consent Form template for non-exempt studies" has been adapted for parental permission. Both forms request signatures from both parents when feasible (per §10 of Doc 1). If only one parent signs, the IRB justification is documented separately.

Doc 2A — Parental Permission Form for Ryan Wang

Study Title: Passive Forearm sEMG/IMU Monitoring During Normal Tennis Practice: A Minimal-Risk Two-Participant Pilot Case Series in Minor Student-Athletes

Principal Investigator / Faculty PI: Dr. YanMing Di, Department of Statistics, Oregon State University, Yanming.Di@oregonstate.edu

Student Researcher / Sensor Participant (your child): Ryan Wang, Westview High School

Technical Assistant: Xiaodong Wang

OSU IRB Approval / Protocol Number: [to be assigned upon approval]

Invitation

Your child Ryan is invited to participate in a minimal-risk research study using passive wearable sensors during his normal tennis practice. Your child is also a student researcher and intended co-author on the engineering manuscript that this work supports. Participation as a sensor participant is voluntary.

Purpose

The purpose of this study is to evaluate whether passive forearm sEMG and IMU motion sensors can support field analysis of muscle-signal and motion trends during normal tennis practice. The study is a two-participant pilot case series and is **not** medical testing or treatment.

What Will Happen

- Small surface EMG electrodes will be placed on the skin over Ryan's forearm to record naturally occurring muscle electrical activity.
- A small IMU motion sensor will be secured on his forearm using a non-invasive strap.

- Ryan will take part in normal tennis practice or practice matches at the California residential court.
- Each session will last up to about 90 minutes including setup and sensor removal.
- Ryan may complete approximately 5 to 10 sessions over the study period.
- He may be asked simple 0–10 ratings for fatigue, exertion, and pain/discomfort.
- No video, audio, heart rate, SpO_2, or other vital-sign data will be collected.

The sensor kit is **passive only**. It records signals; it does not stimulate, deliver medication, deliver heat, or treat any condition. The system is **not a clinical or diagnostic device**, and no medical decisions are made from the data.

Risks

- **Mild skin irritation or redness** from adhesive electrodes or straps.
- **Normal tennis fatigue or soreness** from ordinary practice.
- **Privacy and identifiability risk:** because this is a two-participant case series and Ryan may be named as a co-author, even data labeled with codes may remain indirectly identifiable. Public sharing of raw participant-level data is not permitted under this protocol.
- **Possible perceived pressure to participate** because his father (Xiaodong Wang) is part of the study team. The study team treats this as a serious concern and addresses it through the Independent Assent Administrator (see "Voluntary Participation" below).

Benefits

There is no guaranteed direct benefit to Ryan. He may gain educational experience in wearable sensing, signal processing, and responsible human-subjects research. The study may contribute to knowledge about practical sports-sensing systems.

No medical feedback. Neither you nor Ryan will receive medical interpretation, diagnosis, treatment advice, or individualized health recommendations based on the sensor data.

Compensation and Cost

There is **no compensation** for participation, and **no cost** to Ryan or the family.

Voluntary Participation

Participation is voluntary. Ryan may skip any session and stop at any time without penalty or negative consequences. He does not need to give a reason for stopping. His decision will not affect:

- school standing,
- tennis opportunities,
- family relationships, or

- authorship or scholarly credit for work he has already completed.

To reduce the chance that Ryan feels pressured because of family or co-authorship ties, an **Independent Assent Administrator (IAA)** — an OSU faculty/staff member who is not part of the family or the manuscript — will conduct or witness Ryan's assent discussion privately, without you or any other study-team member present. The IAA confirms that Ryan understands he can decline or withdraw freely.

Confidentiality and Publication

Sensor files will be coded rather than labeled with Ryan's name. The key linking his identity to the code is stored separately on encrypted local or OSU-licensed storage and is accessible only to approved study team members. Because the cohort is N=2 and Ryan may be a named co-author, **complete anonymity cannot be guaranteed**. Publications will avoid unnecessary personal information and will not include identifiable images, video, or audio.

Withdrawal and Data-Removal Limits

Ryan may stop a session, decline a session, or request that data from an incomplete or uncomfortable session not be used. He may also request withdrawal of identifiable or coded data to the extent practicable. **Data that have already been de-identified, aggregated, analyzed, or included in submitted/published results may not be removable.**

Future Use of Data

Data are used only for the approved study and the related publications/presentations described in the IRB application. Materially different future use will require an IRB amendment.

Adult Re-Consent at Age 18

Ryan is expected to turn 18 in September 2026. If data collection continues past his 18th birthday, the study team will pause new data collection for him until written **adult informed consent** is obtained. Pre-18 data remain usable under this parental permission and his minor assent unless he withdraws.

Questions

For questions about the study, contact: Dr. YanMing Di, Yanming.Di@oregonstate.edu, Department of Statistics, Oregon State University, Corvallis, OR 97331.

For questions about research participant rights, contact: Oregon State University Human Research Protection Program / IRB, [HRPP/IRB contact phone and email — to be inserted from current OSU IRB letterhead before submission].

Parent/Guardian Permission Statement

I have read this form and understand the study. I give permission for my child Ryan to participate as a sensor participant in this study. I understand that participation is voluntary and that Ryan may stop at any time without penalty.

Parent/Guardian #1

- Name: _____
- Signature: _____
- Date: _____
- Relationship to Ryan: _____

Parent/Guardian #2 (when feasible)

- Name: _____
- Signature: _____
- Date: _____
- Relationship to Ryan: _____

Child/Participant Name: Ryan Wang

Doc 2B — Parental Permission Form for Theodore (Teddy) Di

Study Title: Passive Forearm sEMG/IMU Monitoring During Normal Tennis Practice: A Minimal-Risk Two-Participant Pilot Case Series in Minor Student-Athletes

Principal Investigator / Faculty PI: Dr. YanMing Di, Department of Statistics, Oregon State University, Yanming.Di@oregonstate.edu

Student Researcher / Sensor Participant (your child): Theodore (Teddy) Di, Crescent Valley High School

Technical Assistant: Xiaodong Wang

OSU IRB Approval / Protocol Number: [to be assigned upon approval]

Special note for Dr. Di as Theodore's parent

Dr. Di is both the principal investigator and Theodore's parent. **Permission is being sought from Dr. Di only in his capacity as Theodore's parent**, separate from his investigator role. The IRB packet

treats this as a conflict of interest and applies the mitigations described in Doc 6, including the Independent Assent Administrator (Doc 8). Dr. Di's authority as PI is not a substitute for parental permission; this signed form is the parental permission record.

Invitation

Your child Theodore is invited to participate in a minimal-risk research study using passive wearable sensors during his normal tennis practice. Your child is also a student researcher and intended co-author on the engineering manuscript that this work supports. Participation as a sensor participant is voluntary.

Purpose

The purpose of this study is to evaluate whether passive forearm sEMG and IMU motion sensors can support field analysis of muscle-signal and motion trends during normal tennis practice. The study is a two-participant pilot case series and is **not** medical testing or treatment.

What Will Happen

- Small surface EMG electrodes will be placed on the skin over Theodore's forearm to record naturally occurring muscle electrical activity.
- A small IMU motion sensor will be secured on his forearm using a non-invasive strap.
- Theodore will take part in normal tennis practice or practice matches at the Corvallis residential court.
- Each session will last up to about 90 minutes including setup and sensor removal.
- Theodore may complete approximately 5 to 10 sessions over the study period.
- He may be asked simple 0–10 ratings for fatigue, exertion, and pain/discomfort.
- No video, audio, heart rate, SpO_2, or other vital-sign data will be collected.

The sensor kit is **passive only**. It records signals; it does not stimulate, deliver medication, deliver heat, or treat any condition. The system is **not a clinical or diagnostic device**, and no medical decisions are made from the data.

Risks

- **Mild skin irritation or redness** from adhesive electrodes or straps.
- **Normal tennis fatigue or soreness** from ordinary practice.
- **Privacy and identifiability risk:** because this is a two-participant case series and Theodore may be named as a co-author, even data labeled with codes may remain indirectly identifiable. Public sharing of raw participant-level data is not permitted under this protocol.

- **Possible perceived pressure to participate** because his father is the principal investigator and is part of the study team. The study team treats this as a serious concern and addresses it through the Independent Assent Administrator (see "Voluntary Participation" below).

Benefits

There is no guaranteed direct benefit to Theodore. He may gain educational experience in wearable sensing, signal processing, and responsible human-subjects research. The study may contribute to knowledge about practical sports-sensing systems.

No medical feedback. Neither you nor Theodore will receive medical interpretation, diagnosis, treatment advice, or individualized health recommendations based on the sensor data.

Compensation and Cost

There is **no compensation** for participation, and **no cost** to Theodore or the family.

Voluntary Participation

Participation is voluntary. Theodore may skip any session and stop at any time without penalty or negative consequences. He does not need to give a reason for stopping. His decision will not affect:

- school standing,
- tennis opportunities,
- family relationships, or
- authorship or scholarly credit for work he has already completed.

To reduce the chance that Theodore feels pressured because his father is the PI, an **Independent Assent Administrator (IAA)** — an OSU faculty/staff member who is not part of the family or the manuscript — will conduct or witness Theodore's assent discussion privately, without his parents or any other study-team member present. The IAA confirms that Theodore understands he can decline or withdraw freely.

Confidentiality and Publication

Sensor files will be coded rather than labeled with Theodore's name. The key linking his identity to the code is stored separately on encrypted local or OSU-licensed storage and is accessible only to approved study team members. Because the cohort is N=2 and Theodore may be a named co-author, **complete anonymity cannot be guaranteed**. Publications will avoid unnecessary personal information and will not include identifiable images, video, or audio.

Withdrawal and Data-Removal Limits

Theodore may stop a session, decline a session, or request that data from an incomplete or uncomfortable session not be used. He may also request withdrawal of identifiable or coded data to the extent practicable. **Data that have already been de-identified, aggregated, analyzed, or included in submitted/published results may not be removable.**

Future Use of Data

Data are used only for the approved study and the related publications/presentations described in the IRB application. Materially different future use will require an IRB amendment.

Adult Re-Consent at Age 18

Theodore is currently 15 and is expected to turn 18 in September 2028. If data collection continues past his 18th birthday, the study team will pause new data collection for him until written **adult informed consent** is obtained. Pre-18 data remain usable under this parental permission and his minor assent unless he withdraws.

Questions

For questions about the study, contact: Dr. YanMing Di, Yanming.Di@oregonstate.edu, Department of Statistics, Oregon State University, Corvallis, OR 97331.

For questions about research participant rights, contact: Oregon State University Human Research Protection Program / IRB, [HRPP/IRB contact phone and email — to be inserted from current OSU IRB letterhead before submission].

Parent/Guardian Permission Statement

I have read this form and understand the study. I give permission for my child Theodore to participate as a sensor participant in this study. I understand that participation is voluntary and that Theodore may stop at any time without penalty.

Parent/Guardian #1

- Name: _____
- Signature: _____
- Date: _____
- Relationship to Theodore: _____

Parent/Guardian #2 (when feasible)

- Name: _____
- Signature: _____
- Date: _____
- Relationship to Theodore: _____

Child/Participant Name: Theodore (Teddy) Di

Document 3 — Minor Assent Forms

OSU's current "Assent Form" template has been adapted for both participants. Both forms are written in plain language; the wording is calibrated for a mature minor reader. The assent discussion is conducted or witnessed **privately** by the Independent Assent Administrator (IAA) without parents or other study-team members present.

Doc 3A — Minor Assent Form for Ryan Wang (age 17)

Study Title: Wearable Muscle and Motion Sensors During Tennis Practice

Researchers: Dr. YanMing Di (Faculty PI); student researcher/co-authors Ryan Wang and Theodore Di

Technical Assistant: Xiaodong Wang

What is this study about?

We are studying how a small wearable sensor can record forearm muscle signals and motion during normal tennis practice. You are also a student researcher and co-author on the engineering paper, but **you still get to decide separately** whether your own body-sensor data can be collected and used.

What will I do?

- Wear small sensor stickers on your forearm.
- Wear a small motion sensor on your forearm.
- Play tennis during normal practice or practice matches at your home court.
- Answer short 0–10 questions about how tired your forearm feels and whether anything hurts.
- Do this for up to about 5 to 10 sessions, each up to about 90 minutes.

Will it hurt?

It should not hurt. The sensors **only record** signals and motion. They do **not** shock, stimulate, treat, or do anything to your muscles. The stickers or strap may cause mild skin redness or irritation. Tennis practice may cause normal tiredness. Tell an adult right away if anything hurts, feels unusual, or if you want to stop.

The system is **not a medical device**, and we will not give you medical advice, diagnosis, training advice, or health recommendations from the data.

Do I have to do this?

No. It is your choice. You can say no now. You can also stop later, skip a session, or stop in the middle of a session. You do not need to give a reason.

You can stop **even if** your parent, the researchers, or co-authors want you to continue.

Saying no — or stopping — will **not** affect:

- school,
- tennis,
- your family,
- your scholarly credit for work you've already done on the engineering paper.

Why are we having this conversation privately?

Because your father is part of the research team, we want to make sure you feel free to say no without anyone in your family being there. The person talking with you right now is the **Independent Assent Administrator** — an OSU staff or faculty member who is not part of your family and not a co-author. They are here to confirm that this is your free choice.

Will my information be private?

The researchers will use a code instead of your name in the sensor files. Because this is a two-person case study and you may be a named co-author on the paper, people may be able to guess that the data are yours. The paper will avoid unnecessary personal information and will not include video or photos of you. The raw data will not be released publicly.

Compensation

There is no payment for being in this study, and no cost to you or your family.

What happens when I turn 18?

You are expected to turn 18 in September 2026. If we are still collecting data after that, we will stop and ask you to give your own adult written consent before continuing. The data we already collected stays usable unless you ask us to stop using it.

Withdrawal limits

If you ask to remove your identifiable or coded data later, we will do that as much as we can. But once data have been de-identified, combined with other data, analyzed, or included in submitted or published results, we may not be able to take them out.

Assent Statement

I understand what will happen. I understand that I can stop at any time and that stopping will not get me into trouble or affect anything else. I agree to participate as a sensor participant in this study.

- Participant Name: Ryan Wang
- Participant Signature: _____
- Date: _____

Independent Assent Administrator (Person Obtaining Assent):

- Name: _____
- Signature: _____
- Date: _____
- Affiliation: OSU faculty/staff (not a parent, not a co-author, not the technical assistant)

Doc 3B — Minor Assent Form for Theodore (Teddy) Di (age 15)

Study Title: Wearable Muscle and Motion Sensors During Tennis Practice

Researchers: Dr. YanMing Di (Faculty PI); student researcher/co-authors Ryan Wang and Theodore Di

Technical Assistant: Xiaodong Wang

What is this study about?

We want to see how a small wearable sensor can record muscle signals and motion from your forearm while you play tennis normally. You are also helping as a student researcher and co-author of the

engineering paper, but **deciding to wear the sensor is a separate choice that is up to you**.

What will I do?

- Wear small sensor stickers on your forearm.
- Wear a small motion sensor on your forearm.
- Play tennis the way you normally do, at your home court in Corvallis.
- Answer short 0–10 questions about how tired your forearm feels and whether anything hurts.
- Do this for about 5 to 10 sessions, each up to about 90 minutes.

Will it hurt?

It should not hurt. The sensors **only record** signals and motion. They do **not** shock you, stimulate your muscles, or treat anything. The stickers or strap might cause mild skin redness or irritation. Tennis itself can make you tired the same way it normally does.

Tell an adult right away if anything hurts, feels weird, or you want to stop.

The system is **not a medical device**, and we will not give you any medical advice, diagnosis, or training advice from what the sensor sees.

Do I have to do this?

No. It is your choice. You can say no right now. You can also stop later, skip a day, or stop in the middle of a session. You do not need to give a reason.

You can stop **even if** your dad, your family, or the researchers want you to keep going.

Saying no — or stopping — will **not** affect:

- school,
- tennis,
- your family,
- the credit you get on the engineering paper for the work you have already done.

Why are we talking privately right now?

Because your dad is the lead researcher, we want to make sure you feel free to say no, without him or anyone in your family in the room. The person talking with you right now is the **Independent Assent Administrator** — an OSU person who is **not** in your family and is **not** a co-author. They are here so this is your free choice.

Will my information be private?

The researchers will use a code instead of your name in the sensor files. Because there are only two people in the study and you may be listed as a co-author of the paper, people who know the paper may be able to guess that the data are yours. The paper will avoid extra personal information about you, and there will be no video, audio, or photos of you. The raw data will not be made public.

Compensation

There is no payment for being in this study, and no cost to you or your family.

What happens when I turn 18?

You are 15 right now. If we are still collecting data when you turn 18 in September 2028, we will stop and ask you to give your own adult written consent before continuing. The data we already collected stays usable unless you ask us to stop using it.

Withdrawal limits

If you ask to remove your identifiable or coded data later, we will do that as much as we can. But once your data have been de-identified, combined, analyzed, or already included in a submitted or published paper, we may not be able to remove them.

Assent Statement

I understand what will happen. I understand that I can stop at any time, and stopping will not get me into trouble. I agree to wear the sensor for this study.

- Participant Name: Theodore (Teddy) Di
- Participant Signature: _____
- Date: _____

Independent Assent Administrator (Person Obtaining Assent):

- Name: _____
- Signature: _____
- Date: _____
- Affiliation: OSU faculty/staff (not a parent, not a co-author, not the technical assistant)

Document 4 — Risk Assessment

Overall expected risk level: Minimal risk. The study uses passive, non-invasive sensors during normal tennis practice. The system does not provide electrical stimulation, treatment, medication, blood draw, radiation, or any invasive procedure.

4.1 Ordinary tennis risk vs. research-added risk

The IRB should distinguish:

- **Ordinary tennis risk** (exists with or without this study): muscle soreness, fatigue, falls, minor injury inherent to tennis itself. The research does not require participants to exceed normal practice intensity.
- **Research-added risk** (exists because of this study): possible mild skin irritation from adhesive electrodes/tape/straps, minor inconvenience from sensor wear, and confidentiality/identifiability risk because the cohort is small and named.

4.2 Risk table

Risk	Description	Likelihood / Severity	Mitigation
Skin irritation / adhesive sensitivity	Adhesive electrodes, tape, or straps may cause temporary redness or irritation.	Low to moderate likelihood; low severity.	Avoid irritated/broken skin; use skin-safe electrodes; remove sensors at any sign of discomfort; check skin after each session; pause/stop if irritation occurs.
Normal tennis fatigue / soreness	Ordinary fatigue/soreness from regular tennis practice.	Moderate likelihood; low to moderate severity.	No added fatigue protocol; rest breaks anytime; stop for pain/unusual discomfort; adult on-court for safety.
Privacy / identifiability (small-cohort)	Physiological and motion data may be sensitive; N=2 case series and named-author status mean	Low likelihood with controls; moderate	Code-based file labeling; identity-key file held separately on encrypted local or OSU-licensed storage; no public release of raw participant-level

Risk	Description	Likelihood / Severity	Mitigation
	identity may be inferred even from "de-identified" data.	severity if breached.	data; no video/audio; minimal personal detail in publications. See Doc 7.
Undue influence / pressure	PI is parent of one minor; technical assistant is parent of the other minor; both minors are intended co-authors.	Low to moderate likelihood; moderate severity if it occurs.	Independent Assent Administrator conducts/witnesses each minor's assent privately (Doc 8); participation–authorship separation written into protocol and consent/assent forms; participant may stop at any time without affecting family, school, tennis, or scholarly credit. See Doc 6.
Court/environment safety	Wet court, heat, poor lighting, or other tennis hazards.	Variable likelihood; low to moderate severity.	Use familiar residential courts only; supervising adult always present; cancel/stop for unsafe weather, wet surface, poor lighting, or other hazard.
Lost device or data breach	Device or storage media containing coded data could be lost or accessed without authorization.	Low likelihood; moderate severity.	Encrypted local or OSU-licensed storage; identity-key file separated; access limited to named team; backup encrypted; reportable as adverse event per Appx C.
Authorship-related pressure	Participant feeling that withdrawing from data collection will hurt his manuscript credit.	Low likelihood; moderate severity.	Participation–authorship separation explicit in protocol, consent, assent; IAA confirms understanding privately; documented as a study protection.
Adverse-event under-reporting	Closed-family team may under-report mild	Low to moderate	Explicit reportable-event list (Appx C); IAA may halt

Risk	Description	Likelihood / Severity	Mitigation
(small, family team)	events because they are normalized.	likelihood; low to moderate severity.	sessions; PI commits to logging and reviewing every event; routine site review at continuing review.

4.3 Stopping rules

A session is stopped if:

- The participant requests to stop.
- Pain, unusual discomfort, dizziness, distress, or concerning fatigue occurs.
- Skin irritation or adhesive reaction occurs.
- Unsafe weather, wet court, poor lighting, or unsafe court conditions occur.
- The on-court adult, the PI, or the IAA determines that continuing is not appropriate.

Document 5 — Sensor and Device Description

This document describes the sEMG/IMU sensor kit used in the study so that OSU IRB reviewers can evaluate the device's risk profile without reference to the engineering manuscript.

5.1 Hardware overview

The system is a low-cost wearable EMG/IMU node that records signals only. It does not stimulate, deliver medication, deliver heat, or perform any therapeutic or diagnostic function.

Component	Specification	Function
MyoWare 2.0 muscle sensor	2000 Hz sampling, 12-bit ADC, ENV (envelope) and RAW outputs	Passive surface EMG recording from forearm muscle group
ICM-20948 IMU	200 Hz, ±2000 dps gyro, ±8 g accelerometer	Passive forearm motion recording (no body stimulation)
ESP32 MCU	BLE-streaming microcontroller	Sample acquisition and wireless data transmission

Component	Specification	Function
Adhesive snap electrodes	Disposable Ag/AgCl, skin-safe	Electrical contact between skin and EMG sensor
Reference electrode	Same type, placed near lateral epicondyle	Reference/ground for the EMG amplifier
Battery	Lithium-polymer, low-voltage	Powers the entire kit. **No mains-electrical connection during use.**

Approximate kit cost: ~$99 in components; the kit is **personally owned** (not OSU lab equipment).

5.2 Signal-only operation

The system is **passive recording only**:

- The MyoWare 2.0 records naturally occurring muscle electrical activity from the skin surface. It does **not** deliver electrical stimulation.
- The ICM-20948 records motion only. It does not interact with the body in any way.
- The ESP32 MCU only streams data over BLE. It does not affect the participant's body.
- The system delivers **no** medication, heat, vibration, current, or therapeutic intervention.

5.3 Software pipeline

Module	Function
`btCentral.py`	BLE central (Python/Bleak) — receives binary TLV frames, writes rotating CSV logs
`capture_loader.py`	Reconstructs hybrid `abs_us` timeline; uses device timestamps within a connection and host receive-time gaps between reconnections
`emg_util_yaml_runner.py`	Per-swing MDF/MNF/RMS feature extraction and reporting
Sampling architecture	ADC DMA at 40 kHz with 10× oversampling; ±25 µs timestamp accuracy at 2000 Hz

Software runs on personal computers controlled by the study team; data is written to encrypted local storage or OSU-licensed storage. See Doc 7.

5.4 Wear, attachment, removal

- Surface EMG electrodes are placed on the skin over the wrist flexor and/or wrist extensor region of the forearm. A reference electrode is placed near the lateral epicondyle.
- The IMU is secured on the forearm with a non-invasive strap.
- The kit is removable at any time; participants may remove sensors themselves.
- Sessions are stopped immediately on discomfort, irritation, dizziness, pain, or request.

5.5 Regulatory framing

- The system is used only for non-clinical research signal recording and engineering analysis.
- Not used for diagnosis, treatment, prevention, monitoring of a medical condition, or any clinical decision-making.
- The study does not test a therapeutic or diagnostic device claim. The PI will address any OSU questions about FDA/device applicability if raised, rather than asserting a categorical "not a medical device" determination upfront.
- No medical interpretation, diagnosis, training advice, or individualized health recommendation is provided to participants or parents based on the data.

Document 6 — Conflict of Interest Management Plan

6.1 Disclosed conflicts

The study team contains two structurally significant family relationships:

1. **Dr. YanMing Di (PI) is the parent of minor participant Theodore Di.**
2. **Xiaodong Wang (technical assistant) is the parent of minor participant Ryan Wang.**

Both minors are also intended student co-authors on the engineering manuscript supported by this study. This creates symmetric undue-influence risk and potential publication-interest bias.

There are **no financial conflicts** disclosed under this protocol. The hardware is personally owned, no external sponsorship is involved, and no investigator will gain financial benefit from study outcomes beyond ordinary academic credit.

6.2 Mitigations

#	Mitigation	Where it appears
M1	**Independent Assent Administrator (IAA)** — an OSU faculty/staff member who is not a parent, not a co-author, not the technical assistant, and has no publication interest. The IAA conducts or witnesses each minor's assent privately, without parents or other study-team members present.	Doc 8
M2	**Participation–authorship separation** — withdrawal from data collection does not affect family relationships, school standing, tennis opportunities, or appropriate scholarly credit for work already completed. The decision to participate is documented separately from any manuscript authorship agreement.	Doc 1 §10, §18; Doc 2A/2B; Doc 3A/3B
M3	**Dr. Di does not conduct his own son's assent.** Dr. Di's role with respect to Theodore is parental permission only. Investigator-style discussions with Theodore about participation are not conducted by Dr. Di alone.	Doc 1 §10, §22; Doc 8
M4	**Xiaodong Wang does not conduct his own son's assent.** The technical assistant's role with respect to Ryan is parental permission only and on-court technical/safety support.	Doc 1 §10, §22; Doc 8
M5	**Manuscript COI declaration** — the eventual IEEE manuscript will explicitly disclose the family relationships, the IAA mitigation, and the documented parental permission.	Appx A
M6	**OSU Beaver Research Hub disclosure** — Dr. Di will file the appropriate COI disclosure through OSU's Beaver Research Hub. The PI will confirm with the OSU COI Office whether familial-participant disclosures require additional management plan review.	Cayuse submission
M7	**Adverse-event reporting parity** — to counter possible under-reporting in a closed-family team, every reportable event (Appx C) is logged, reviewed by the PI, and surfaced at OSU continuing review.	Appx C

6.3 Reporting and amendments

Any new conflict of interest, including new financial interests, new familial relationships, or any change to the study-team structure, is reported to OSU through Cayuse HE as a protocol amendment **before** taking effect.

6.4 Acknowledgment

By signing the IRB submission cover page, Dr. YanMing Di and Xiaodong Wang acknowledge their respective familial relationships with the minor participants, the specific mitigations above, and their commitment to follow this management plan throughout the study.

- Dr. YanMing Di (PI; parent of Theodore Di) — Signature: _____ Date: _____
- Xiaodong Wang (technical assistant; parent of Ryan Wang) — Signature: _____ Date: _____

Document 7 — Data Security Plan

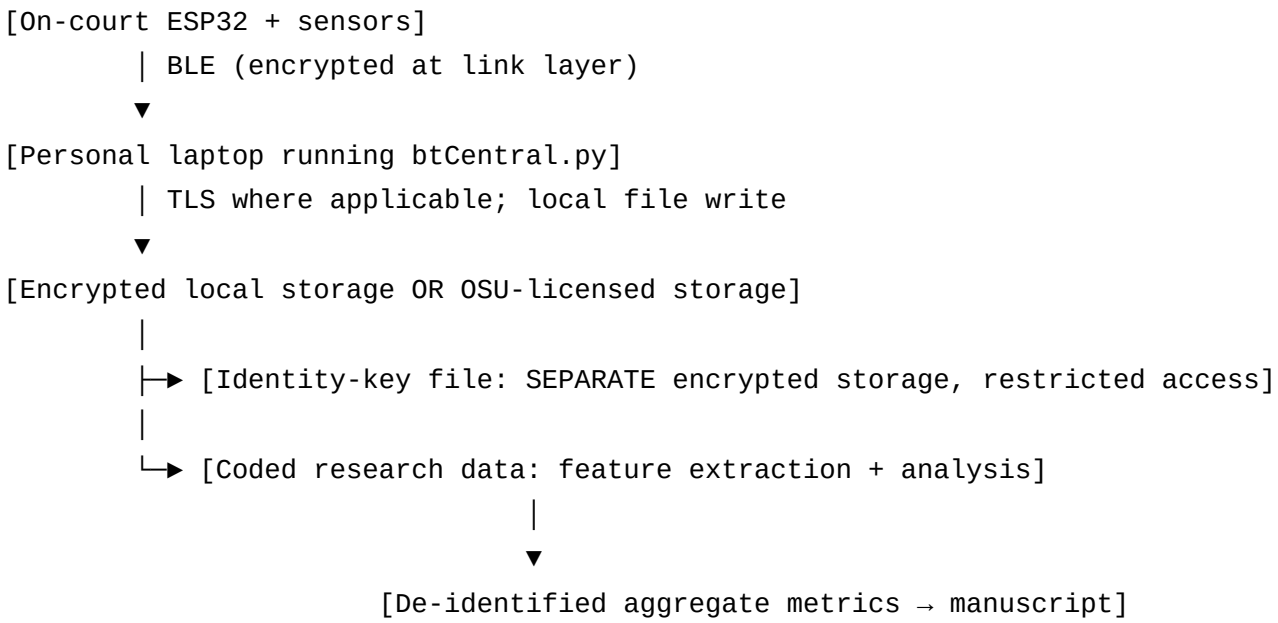
7.1 Data classification

Per OSU's data-classification framework, the data collected under this protocol are anticipated **Level 2 (Confidential)**: identifiable or coded data with minimal confidentiality risk if breached. Level 3 (High) does not apply because no PHI, financial, or regulated identifiers are collected.

7.2 N=2 indirect-identifiability acknowledgment

Because the cohort is N=2 and both participants may be named as manuscript authors, **raw and coded physiological data are treated as potentially identifiable even after removal of direct identifiers**. Public sharing of raw participant-level data is **not** permitted under this protocol, and any IEEE data-availability statement reflects this constraint.

7.3 Data flow



7.4 Storage controls

Control	Implementation
Storage location	Encrypted local drive on study-team computer, **or** OSU-licensed cloud storage (e.g., OSU OneDrive/Box). **No** non-OSU-licensed cloud storage.
Encryption at rest	Full-disk encryption (BitLocker, FileVault, or LUKS) on study computers. Identity-key file additionally encrypted with a separate passphrase.
Encryption in transit	BLE link-layer encryption between sensor and laptop; TLS for any cloud sync.
Access control	Limited to named study-team members; OSU-licensed systems use OSU SSO; local storage protected by OS user accounts.
Identity-key separation	Identity-to-code mapping stored in a separate encrypted file from the research data.
OS hygiene	Patched OS; current antivirus; software firewall enabled; disk-level encryption enabled.
Backup	Routine encrypted backup; backups follow the same access and encryption controls.

7.5 Retention and destruction

- **Retention period:** coded research data retained for **[N] years** (final value TBD by PI; e.g., 5 years) after publication or study closure.
- **Identity-key destruction:** the identity-to-code mapping is destroyed earlier, when no longer needed for the manuscript or any IRB-approved follow-up.
- **De-identified aggregate metrics:** may be retained longer with IRB approval if needed for academic recordkeeping.
- **Disposal:** secure cryptographic erasure at end of retention.

7.6 Sharing

- **Raw data:** not publicly shared. Not provided to non-team members without IRB amendment.
- **Manuscript-grade summary metrics:** may be shared with IEEE editors/reviewers under journal confidentiality, and with the corresponding author on reasonable request, subject to the IRB-approved data-sharing limits.

7.7 Breach reporting

A lost device, lost storage media, unauthorized access, or any other breach of the data security plan is reported as an adverse event per Appx C. OSU HRPP and (if Level 2 cloud storage was involved) OSU Information Security are notified per OSU policy.

Document 8 — Independent Assent Administrator (IAA) Designation and Script

8.1 Designation

The Independent Assent Administrator for this study is:

- **Name:** [TBD — OSU faculty/staff designee, to be named by PI before submission]
- **Title:** [TBD]
- **Department:** [TBD]
- **Email:** [TBD]
- **Phone:** [TBD]

Required qualifications

The IAA must:

1. Be an OSU faculty or staff member with current CITI Group 1 or Group 2 — Human Subjects Protection training.
2. **Not** be a parent of either minor participant.
3. **Not** be a co-author of the manuscript supported by this study.
4. **Not** be the technical assistant.
5. **Not** have a publication, financial, or supervisory interest in the study.
6. Have time and willingness to conduct or witness assent discussions privately with each minor at study entry, and to be available for re-affirmation or withdrawal discussions during the study.

Designation acknowledgment

I have read this designation and agree to serve as the Independent Assent Administrator for this study under the conditions above.

- IAA Name: _____
- Signature: _____
- Date: _____

8.2 IAA assent script

The IAA conducts or witnesses each minor's assent discussion **privately** — without parents, the PI, the technical assistant, or any co-author present. The script below is calibrated for a mature minor reader; the IAA may rephrase as needed for a 15-year-old (Teddy) or 17-year-old (Ryan) without altering substance.

[Step 1 — Introduce role and purpose]

> Hi [Ryan / Teddy], I'm [Name], from [Department] at Oregon State University. I'm not in your family, I'm not a researcher on this study, and I'm not a co-author on the paper. The reason we're talking privately is to make sure that **your decision** to be in the study is yours alone — not because anyone in your family or anyone else is hoping you'll agree.

[Step 2 — Confirm comprehension of the study]

> In your own words, what do you think this study is about? What will you do if you say yes?

(IAA listens; clarifies where understanding is incomplete using the assent form.)

[Step 3 — Voluntariness]

> Saying no right now is completely fine. Saying yes now and then changing your mind later is also fine. You can stop in the middle of any session. You don't need to give a reason.

[Step 4 — No consequences for declining]

> Saying no — or stopping later — will not affect:
>
> - your school,
> - tennis,
> - your family,
> - or the credit you get on the engineering paper for the work you've already done.
>
> Are you sure you understand that? Anything you want to ask?

[Step 5 — No medical advice]

> The sensor only records signals. It is not a medical device. No one is going to give you medical advice, training advice, or diagnosis from the data. If you ever feel uncomfortable about how the data are used, tell me — not just your dad or the researchers.

[Step 6 — Stop, withdraw]

> If you ever want to stop, you can tell anyone — me, your dad, the researchers, or any adult on court. The session stops right then. We can also remove your data later, with the limits that are written in the form.

[Step 7 — Final assent question]

> Knowing all of this, do you want to participate?

(If yes, the IAA proceeds to signature on Doc 3A or 3B and signs the Person-Obtaining-Assent line. If no — or if uncertain — the IAA documents the decision, the participant is not enrolled, and there is no consequence to family or school.)

8.3 Re-affirmation

The IAA is available for short re-affirmation discussions during the study at any time the participant, the IAA, the PI, or any adult on court raises a concern. Re-affirmation may be triggered by visible discomfort, a question from the participant about what the data will be used for, or any indication that the participant feels pressured to continue.

If the IAA halts a session under §19 of Doc 1, the halt is documented and the IRB is informed per Appx C if it constitutes a deviation or adverse event.

Appendix A — Manuscript Ethics Statement (IEEE-ready)

The following paragraphs, drawn from OSU IRB approval and this protocol, are intended for direct use in the eventual IEEE manuscript. Bracketed values are filled in after IRB approval.

A.1 Ethics statement (Methods section)

> The study was reviewed and approved by the Oregon State University Human Research Protection Program / Institutional Review Board (protocol #[approval number]; approval date [date]). The study was conducted in accordance with the U.S. Common Rule (45 CFR 46), 45 CFR 46 Subpart D — Additional Protections for Children, and the Declaration of Helsinki. Both participants are minors. Written parental permission and written minor assent were obtained before any data collection. Minor assent was obtained through an Independent Assent Administrator who was not a parent, co-author, or other team member with publication interest in the study.

A.2 Conflict of Interest declaration (end-matter)

> The authors disclose that one minor participant/co-author (Theodore Di) is the child of the principal investigator and corresponding author (Dr. YanMing Di). The second minor participant/co-author (Ryan Wang) is the child of the technical assistant (Xiaodong Wang), who is acknowledged in the manuscript and is not an author. To mitigate potential undue influence, minor assent for both participants was obtained through an Independent Assent Administrator (an Oregon State University faculty/staff member with no publication interest in the study), and parental permission was documented before any data were collected. The authors have no financial conflicts of interest.

A.3 Data availability (end-matter)

> Raw, participant-level data are not publicly available because the IRB-approved protocol does not authorize public sharing of identifiable or coded physiological data from this two-participant case series, where indirect identifiability persists due to the small cohort size and the named-author status of the participants. De-identified aggregate summary metrics are available from the corresponding author on reasonable request and subject to the IRB-approved data-sharing limits.

A.4 Funding (end-matter)

> This work received no external funding. The hardware kit was personally owned by the study team. Oregon State University provided IRB sponsorship and scientific oversight via the principal investigator's appointment but did not provide laboratory facilities, equipment, or financial support.

A.5 Author contributions

Author	Contributions
Ryan Wang	Sensor system co-design (hardware/software); primary data collection at the California site; analysis pipeline; manuscript drafting.
Theodore (Teddy) Di	Sensor system co-design (hardware/software); second-site data collection (Corvallis); cross-site reproducibility analysis; manuscript review and revision.
Dr. YanMing Di	Study conception; design guidance for sensor system, signal-processing algorithms, and mathematical modeling; statistical and scientific supervision; manuscript review; corresponding author.

The technical assistant (Xiaodong Wang) contributed on-court technical support, sensor setup assistance, and equipment custody/maintenance, and is recognized in the Acknowledgments.

A.6 AI-disclosure statement (Acknowledgments — include only if AI-assisted drafting was used)

> AI-assisted drafting tools were used to assist with organization and language editing of [specify sections]. All study design, data collection, analysis, interpretation, and final manuscript content were reviewed and approved by the authors.

A.7 Acknowledgments

> The authors thank Xiaodong Wang for assistance with hardware setup, on-court technical support, and equipment custody during the data-collection sessions. The authors also thank [IAA name and title] for serving as the Independent Assent Administrator for this study.

Appendix B — Recruitment Plan (Closed Cohort)

B.1 Cohort

The study cohort is **closed at N=2** and consists of:

1. Ryan Wang, Westview High School (California), age 17 at start.
2. Theodore (Teddy) Di, Crescent Valley High School (Corvallis, OR), age 15 at start.

Both minors are known to the study team in advance of enrollment as student researcher/co-authors on the engineering manuscript that this study supports.

B.2 No public recruitment

There is **no** public recruitment. There are **no** flyers, advertisements, social-media posts, school-based announcements, coach-mediated invitations, or any other recruitment channels. Neither Westview HS nor Crescent Valley HS is involved as a research site, recruitment site, or sponsor.

B.3 Enrollment sequence

1. OSU IRB approval is obtained.
2. The IAA is designated and accepts.
3. Parental permission (Doc 2A/2B) is sought from both parents of each minor, when feasible.
4. Once parental permission is received, the IAA conducts/witnesses each minor's assent discussion privately and obtains written assent (Doc 3A/3B).
5. Data collection begins only after both written parental permission and written minor assent are on file.

B.4 No additional minors

No additional minors will be added to this study under this protocol. Adding a participant would require an IRB amendment and a new consent/assent and risk reassessment.

Appendix C — Adverse Event and Continuing Review Procedure

C.1 Definitions

- **Adverse event (AE):** any untoward physical, psychological, or social occurrence experienced by a participant during or after a study session, whether or not the study team believes the event was caused by the research.
- **Protocol deviation:** any departure from the IRB-approved protocol, intentional or unintentional.
- **Reportable new information (RNI):** any AE, deviation, or other event that meets OSU HRPP definitions of unanticipated problems involving risks to participants or others.

C.2 Reportable events (non-exhaustive list)

- Skin redness lasting beyond a session, rash, or blistering from electrodes/tape/straps.
- Pain or unusual discomfort from device placement.
- Privacy breach, lost device, or lost data.
- Unauthorized access to study storage.
- Distress or perceived pressure to participate (whether reported by the participant, the IAA, or another team member).
- Any tennis injury occurring during a research-recorded session — even if likely ordinary sport-related — is logged and evaluated.
- Any IAA-initiated session halt under §19 of Doc 1.

C.3 Reporting timeline

- **Serious AE or unanticipated problem:** reported to OSU HRPP through Cayuse HE within the OSU-required timeline (typically 5 business days; PI verifies current value at submission).
- **Routine AE / minor deviation:** logged in the study record and summarized at continuing review.
- **Privacy/data breach:** immediately to OSU HRPP and, if Level 2 cloud storage was involved, OSU Information Security.

C.4 Continuing review and amendments

- Continuing review per OSU HRPP timeline as stated in the IRB approval letter (typically annual for non-exempt studies).

- Amendments for any material change to procedures, study team, sites, sensor configuration, or risk profile are submitted **before** taking effect.
- Adult-consent transition for each participant is submitted as an amendment if data collection is anticipated to continue past the 18th birthday.

C.5 Records retained

- AE log: date, description, response, outcome, reporter.
- Deviation log: same fields.
- IAA contact log: any session halts, re-affirmation discussions, or participant-reported concerns.

Appendix D — CITI Training Tracker

All study team members must complete CITI **Group 1 or Group 2 — Human Subjects Protection** before submission, and renew every 3 years. Responsible Conduct of Research is **not** a substitute. Non-OSU team members may complete OSU CITI with OSU affiliation added, or submit equivalent documentation acceptable to the IRB.

Person	Role	Required course	Status	Completion date	Renewal due	CITI email matches Banner email?
Dr. YanMing Di	PI	Human Subjects Protection (Group 1 or 2)	Pending	[TBD]	+3 years	Yes (OSU email)
Ryan Wang	Student researcher / co-author	Human Subjects Protection (Group 1 or 2), via OSU affiliation	Pending	[TBD]	+3 years	N/A (non-OSU)

Person	Role	Required course	Status	Completion date	Renewal due	CITI email matches Banner email?
Theodore (Teddy) Di	Student researcher / co-author	Human Subjects Protection (Group 1 or 2), via OSU affiliation	Pending	[TBD]	+3 years	N/A (non-OSU)
Xiaodong Wang	Technical assistant	Human Subjects Protection (Group 1 or 2), via OSU affiliation or equivalent	Pending	[TBD]	+3 years	N/A (non-OSU)
[IAA name]	Independent Assent Administrator	Human Subjects Protection (Group 1 or 2)	[TBD]	[TBD]	+3 years	Yes (OSU email)

Appendix E — IEEE Compliance Checklist

The following items are tracked from study design through manuscript submission to satisfy IEEE publication-ethics requirements (see `IRB_requirements.md` Part 2).

#	Item	Source	Status
E1	IRB approval reference (institution + protocol #) included in Methods	IEEE Submission and Peer Review Policies	Pending OSU approval

#	Item	Source	Status
E2	Consent statement (parental permission + minor assent) included in Methods	IEEE Submission and Peer Review Policies	Drafted (Appx A.1)
E3	Declaration of Helsinki conformance statement included in Methods	Biomedical-venue norm	Drafted (Appx A.1)
E4	Conflict of Interest declaration included in end-matter	IEEE Submission and Peer Review Policies	Drafted (Appx A.2)
E5	Data availability statement aligned with IRB-approved data plan	Biomedical-venue norm	Drafted (Appx A.3)
E6	Funding statement included in end-matter	Biomedical-venue norm	Drafted (Appx A.4)
E7	Author Contributions section meeting IEEE authorship criteria	IEEE Ethical Requirements	Drafted (Appx A.5)
E8	AI-disclosure paragraph in Acknowledgments (only if AI-assisted drafting used)	IEEE Submission and Peer Review Policies	Conditional (Appx A.6)
E9	Acknowledgment of non-author contributors (Xiaodong Wang, IAA)	IEEE Ethical Requirements	Drafted (Appx A.7)
E10	ORCID for each author (compulsory for IEEE Sensors Journal; recommended elsewhere)	Biomedical-venue norm	Pending
E11	Original-work statement; not under simultaneous review elsewhere	IEEE Submission and Peer Review Policies	Author confirmation at submission
E12	If posting a preprint, include IEEE preprint boilerplate	IEEE Submission and Peer Review Policies	Conditional

End of IRB Application Packet (Draft v0.1).